Logistic Regression

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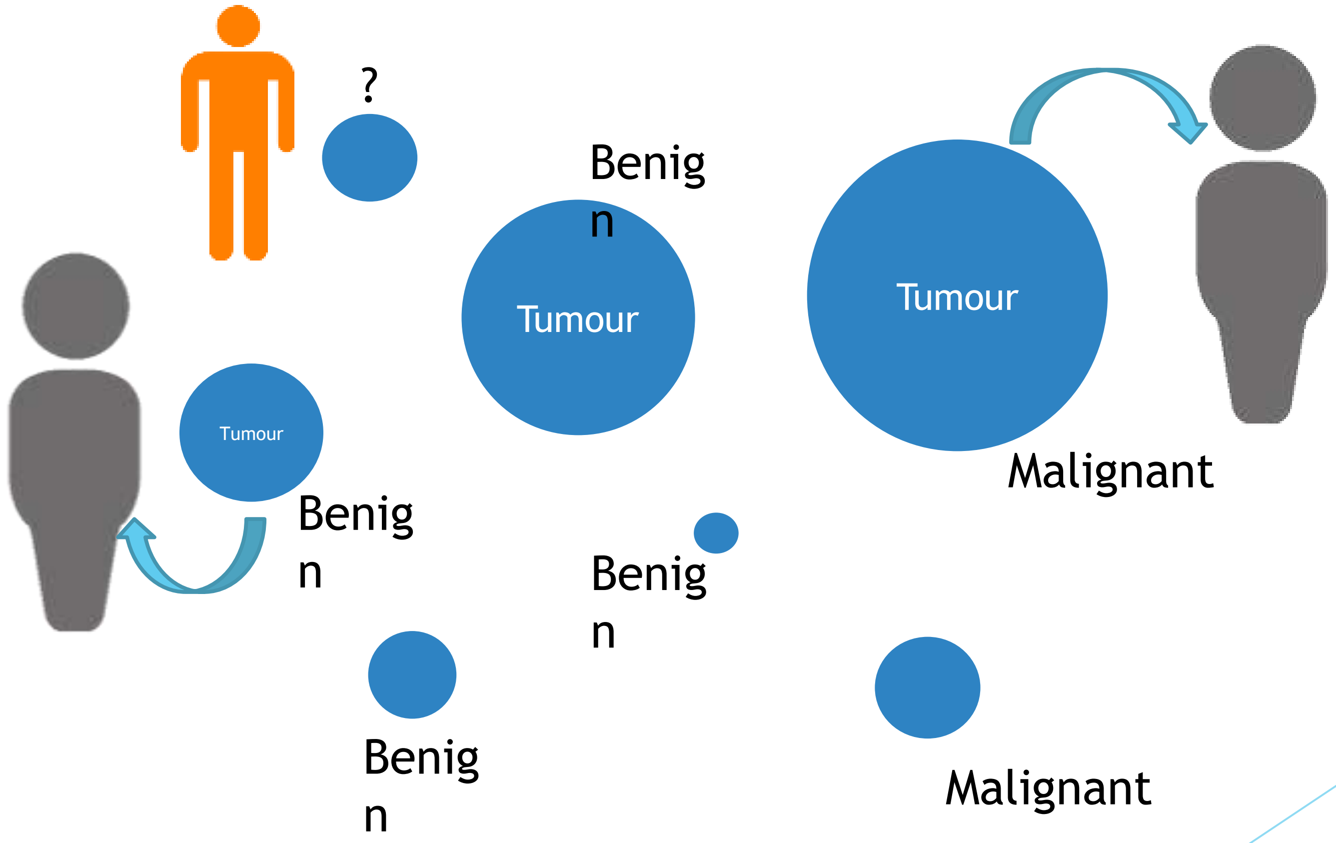
Agenda

- Problem Statement
- Linear Regression vs Logistic Regression
- Sigmoid Function
- Log Odds Ratio
- How to find parameters
- Classification Metrics

Example 1: Tumour Size Vs Tumour Type



Example: Tumour Size Vs Tumour Type

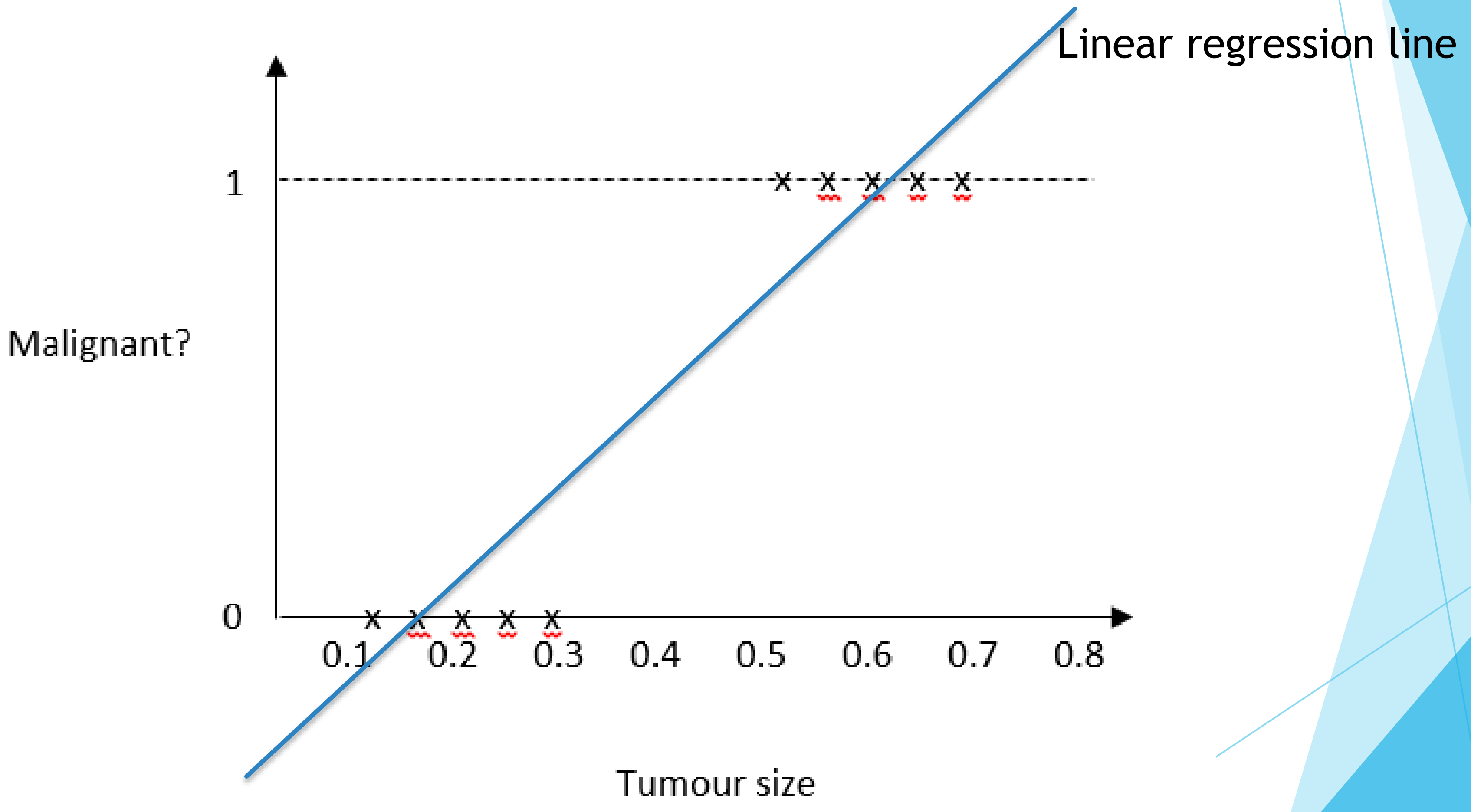
The Data Set

Tumour Size	Tumour Type
0.5	0
2.5	0
0.75	1
5	0
15	1
⋮	⋮

0 = Benign ⇐ Absence of Malignance

1 = Malignant ⇐ Presence of Malignance

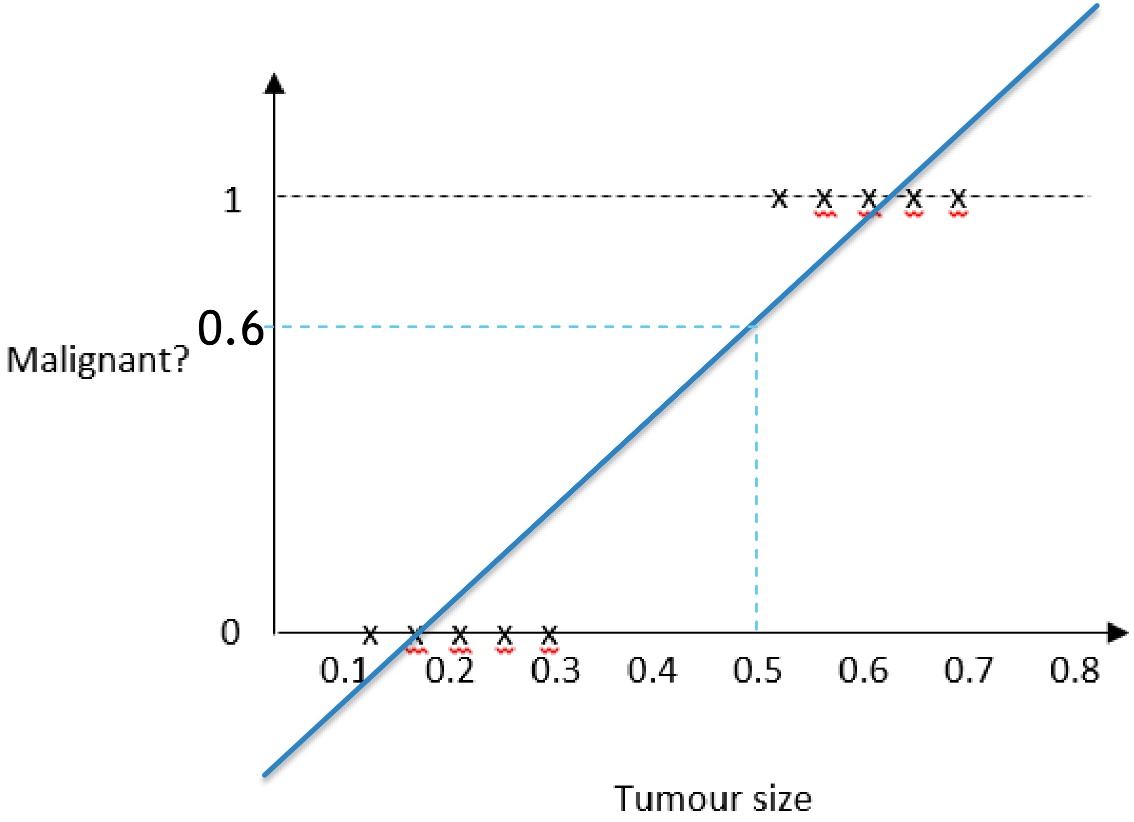
Can we use Linear Regression to Model the Problem?

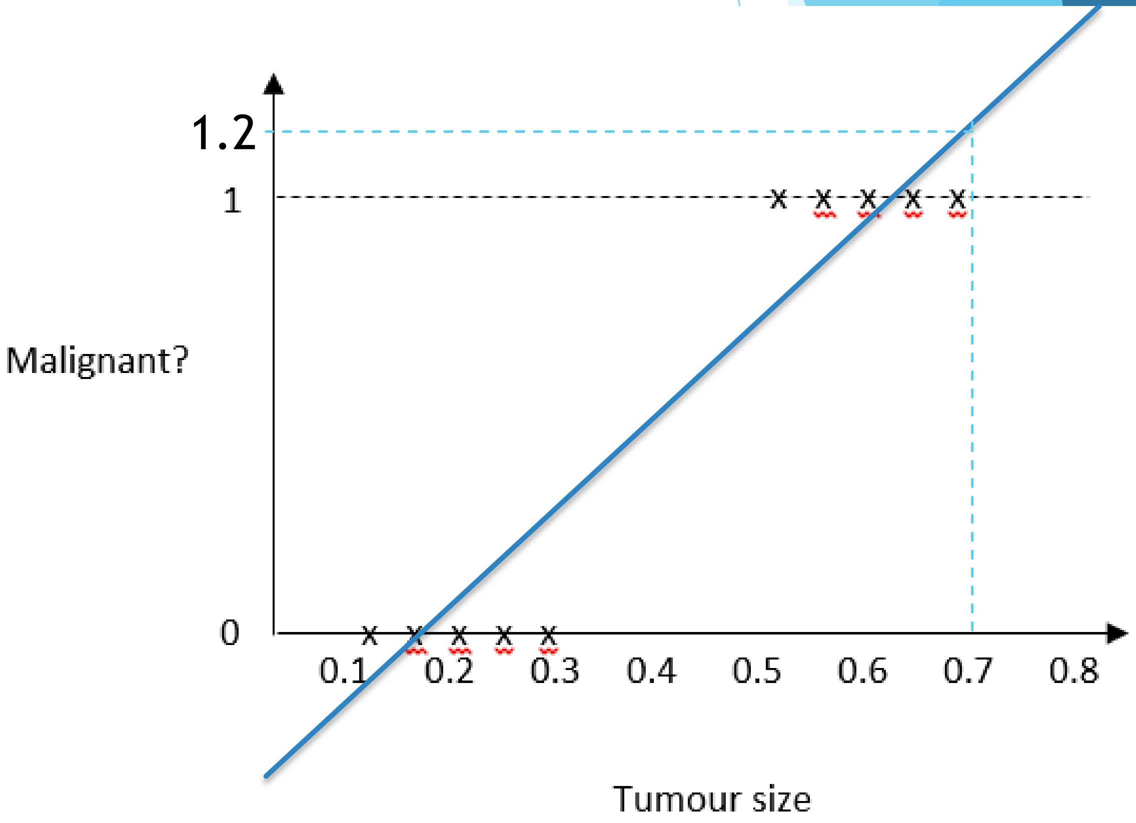


Limitations???

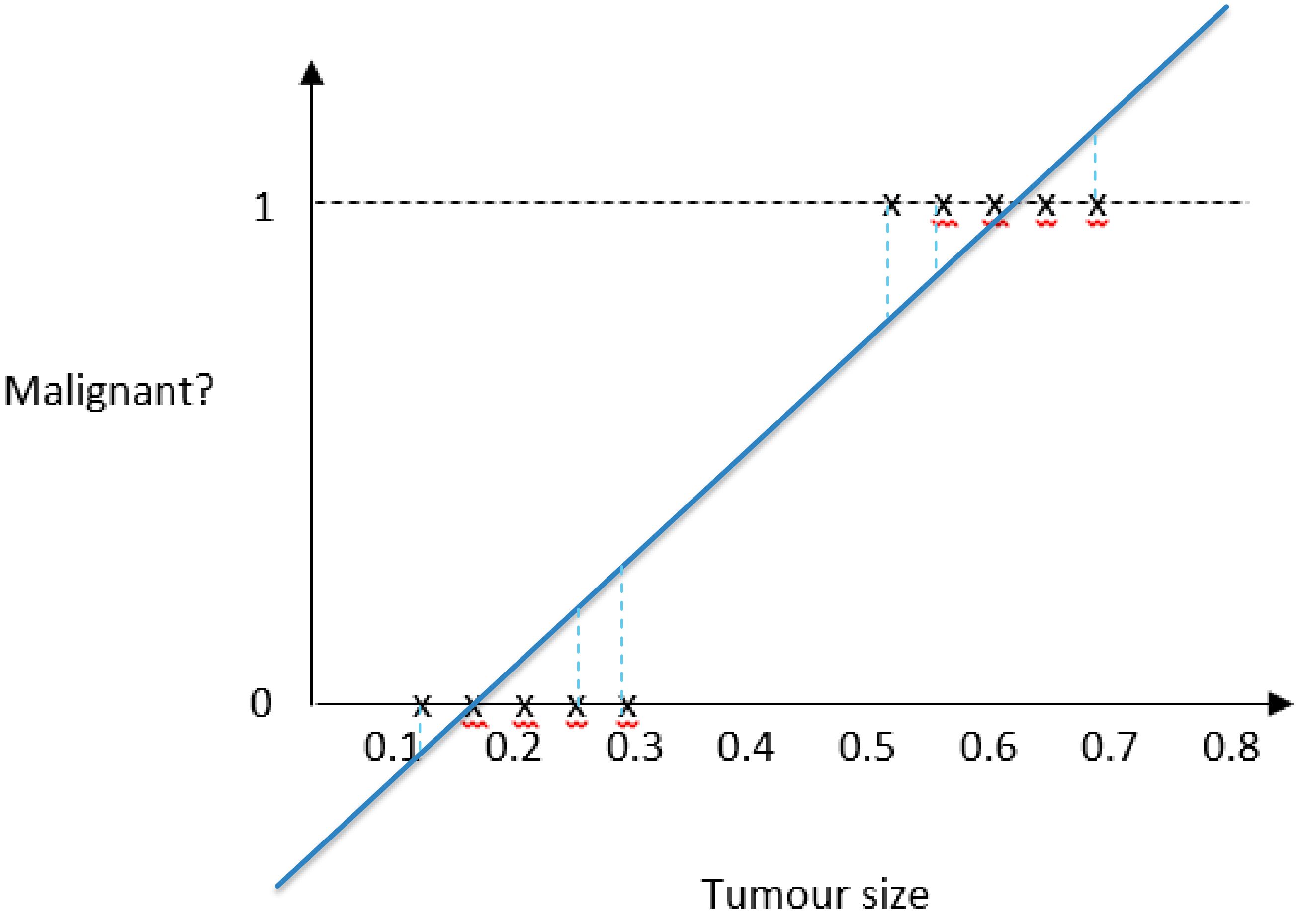
What do we mean by tumour type = 0.6?!!

What do we mean by tumour type = 1.2?!!



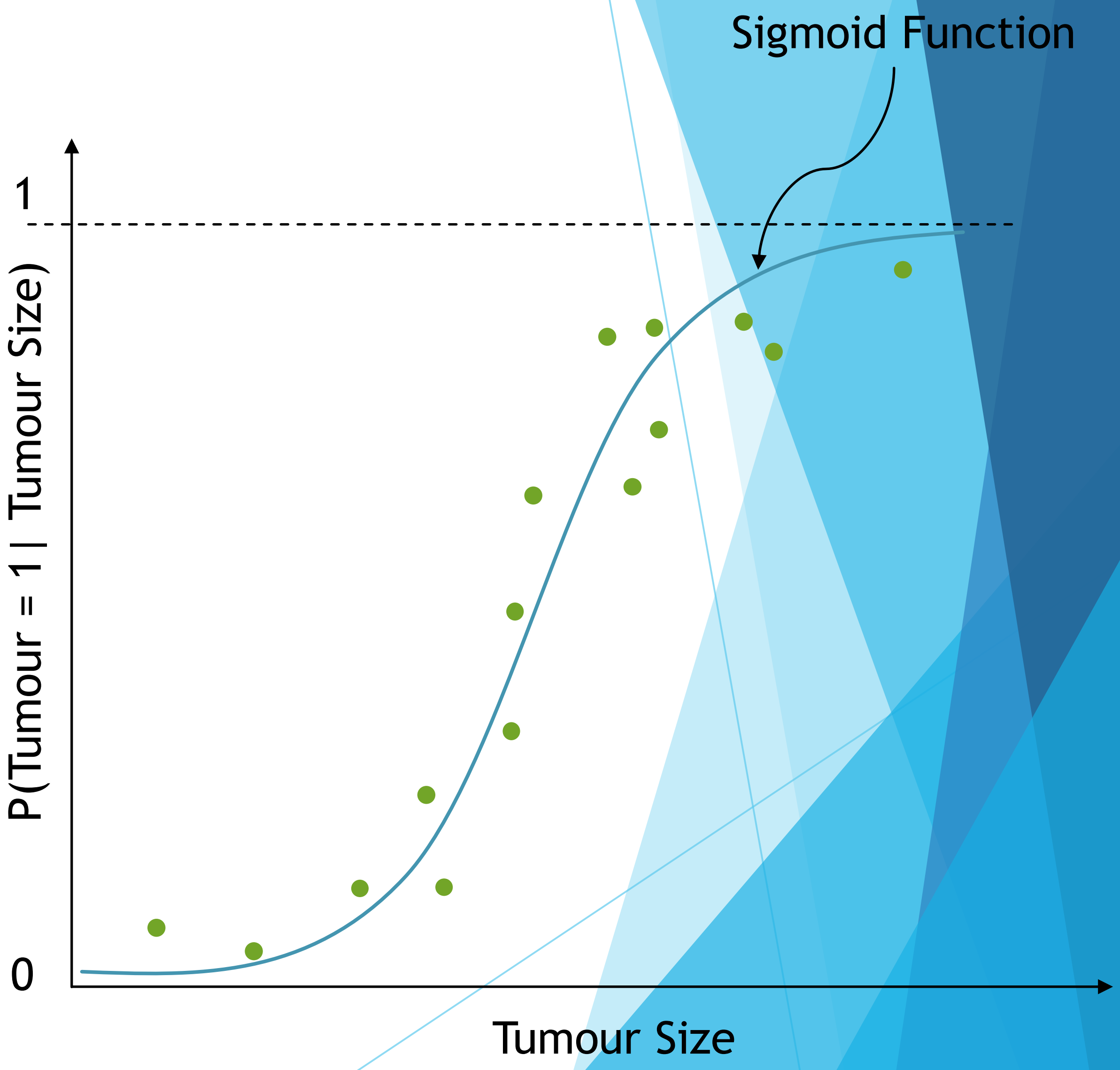


Huge RSS!!

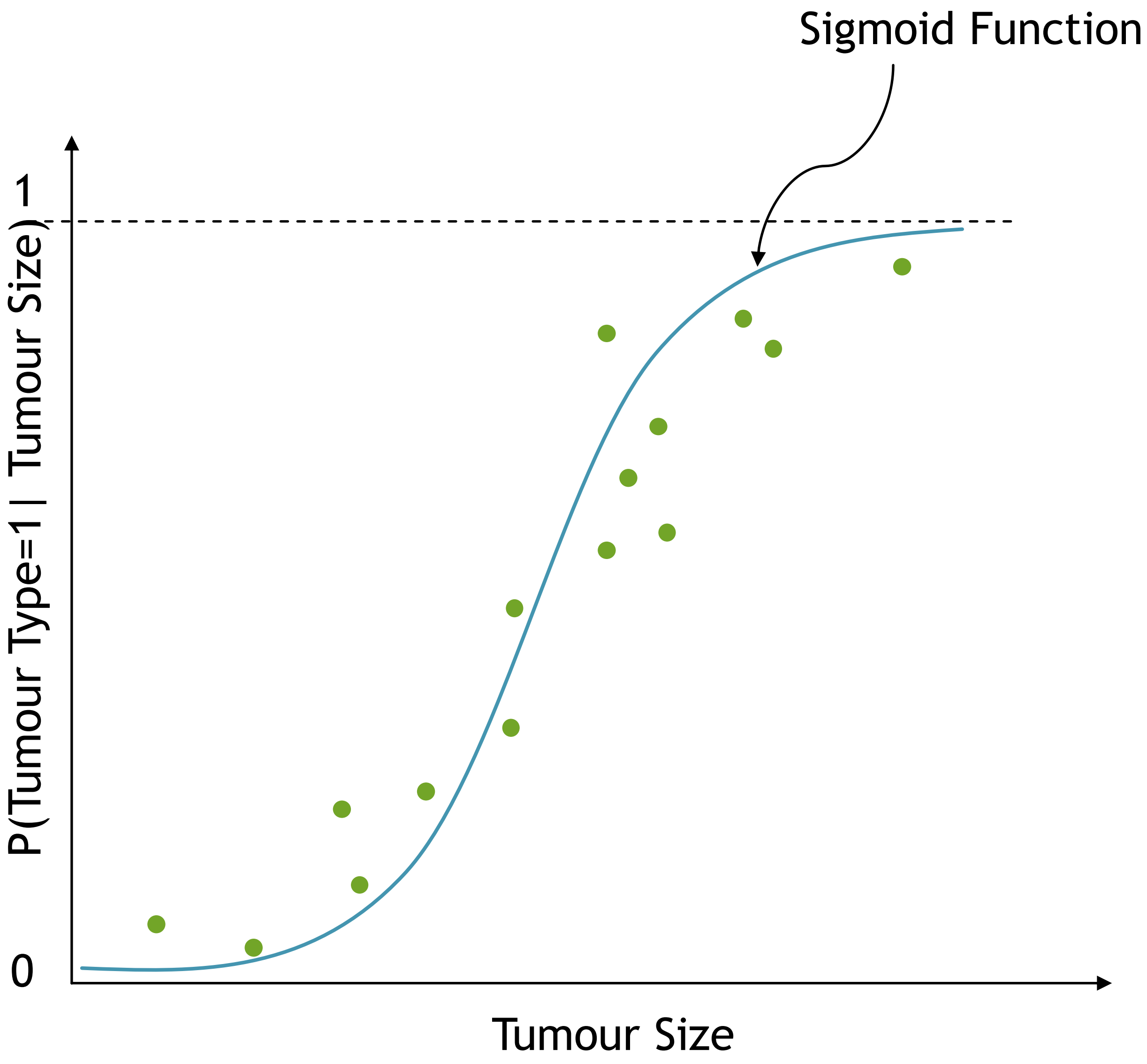


Probabilistic Outcome

Tumour Size	P(Tumour = 1| Tumour Size)
0.0 - 5.0	0.05
5.0 - 10.0	0.25
10.0 - 15.0	0.27
15.0 - 20.0	0.38
25.0 - 30.0	0.43
⋮	⋮



Sigmoid Function



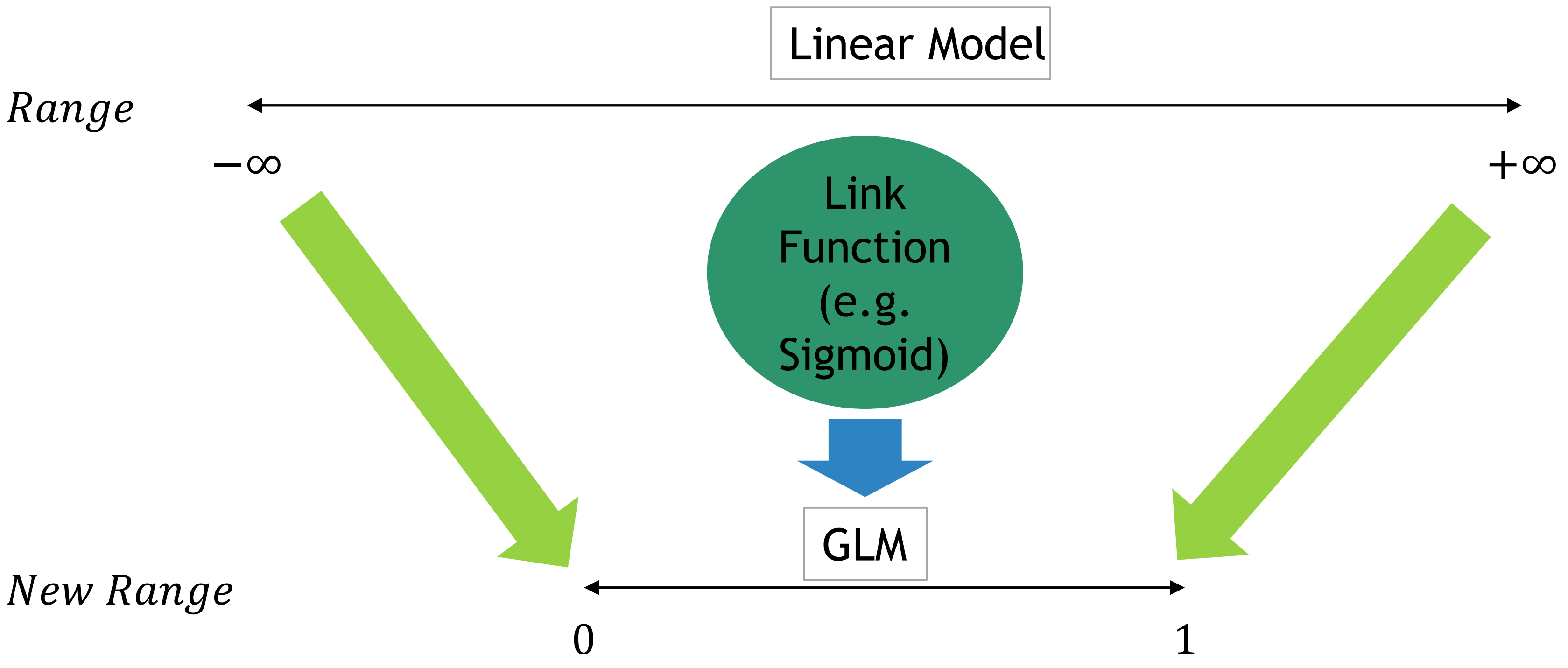
$$g(z) = \frac{1}{1 + e^{-z}}$$

In this Problem: z = a + b*x

Therefore, our logistic regression model is:

$$\mathrm{P}(\mathrm{Y} = 1 | \mathrm{X} = \mathrm{x}) = g(z) = g(a + b\ x) = \frac{1}{1 + e^{-(a+bx)}}$$

GLM



When you take a linear model $y = a + b_1x_1 + b_2x_2 + \ldots + b_dx_d$ *that ranges from* $-\infty$ *to* $+\infty$ *and squeeze it into the range* $(0, 1)$ *using a link function then what you are building is called Generalized Linear Model.*

Logistic Regression Model

$$\underbrace{P(Y = 1|X = x) = \frac{1}{1+ e^{-(a+bx)}}}_{\text{Non-Linear Function!}}$$

Conditional Probability of Y given that X takes the value x,
e.g. Probability that a tumour type (Y) is malignant (Y=1) given that its size (X) is 7 mm (x)

Range of Sigmoid Function

$$g(z) = \frac{1}{1 + e^{-z}}$$

When z = -Inf,

$$g(z) = \frac{1}{1 + e^{inf}} = 0$$

When z = inf,

$$g(z) = \frac{1}{1 + e^{-inf}} = \frac{1}{1} = 1\ (limit)$$

Therefore, $0 < g(z) < 1$

Note: $\exp(-Inf) = 0$

$\exp(Inf) = Inf$

Linear Transformation

$$P(Y = 1|\ x) = g(a + bx) = \frac{1}{1+ e^{-(a+bx)}}$$

Let us write P(Y=1|x) as p(x), for simplicity

$$\Rightarrow\ p(x) = \frac{1}{1 +\ e^{-(a+bx)}}$$

$$\Rightarrow \frac{1}{p(x)} = 1 +\ e^{-(a+bx)}$$, Taking Reciprocal

$$\Rightarrow \frac{1}{p(x)} - 1 = e^{-(a+bx)}$$

$$\Longrightarrow \frac{1 - p(x)}{p(x)} = e^{-(a+bx)}$$

$$\Longrightarrow \frac{p(x)}{1 - p(x)} = e^{(a+bx)}$$

$$\Longrightarrow \underbrace{log\left(\frac{p(x)}{1 - p(x)}\right) = a + bx}_{\text{Linear model}}$$

Logistic Regression Model in Linear Form

$$\Rightarrow log\left(\frac{p(x)}{1-p(x)}\right) = a + bx$$

$$\Rightarrow log\left(\frac{P(Y=1|X)}{1-P(Y=1|X)}\right) = a + bx$$

Odds of Y=1

Interpreting the coefficient b:

With one unit increase in the value of x the log of odds of Y will increase by an amount equal to b, provided all the other factors remains constant.

Odds Vs Probability

$$\text{Odds (in favour)} = \frac{P(Success)}{P(Failure)} = \frac{P(Success)}{1 - P(Success)}$$

Note: Probability ranges between 0 and 1, while odds ranges between 0 and infinity

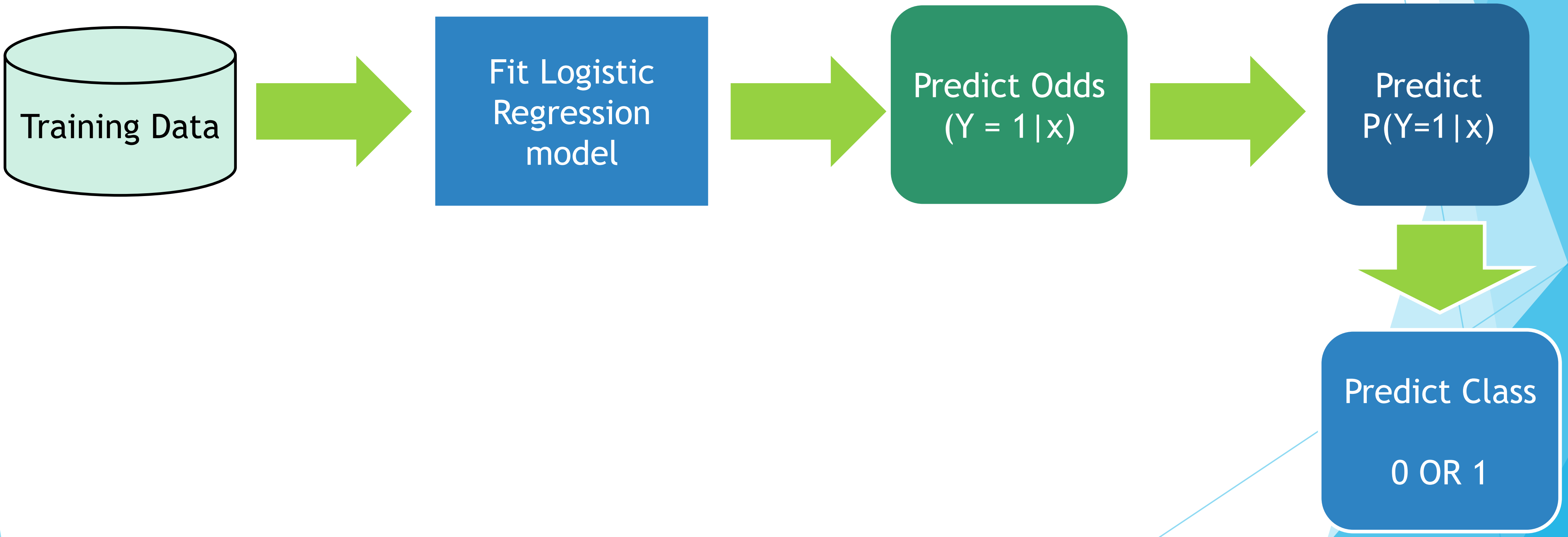
How this Entire Process Work?

x = Tumour Size
Y = Tumour Type (0/1)

$$log\left(\frac{p(x)}{1-p(x)}\right) = a + bx$$

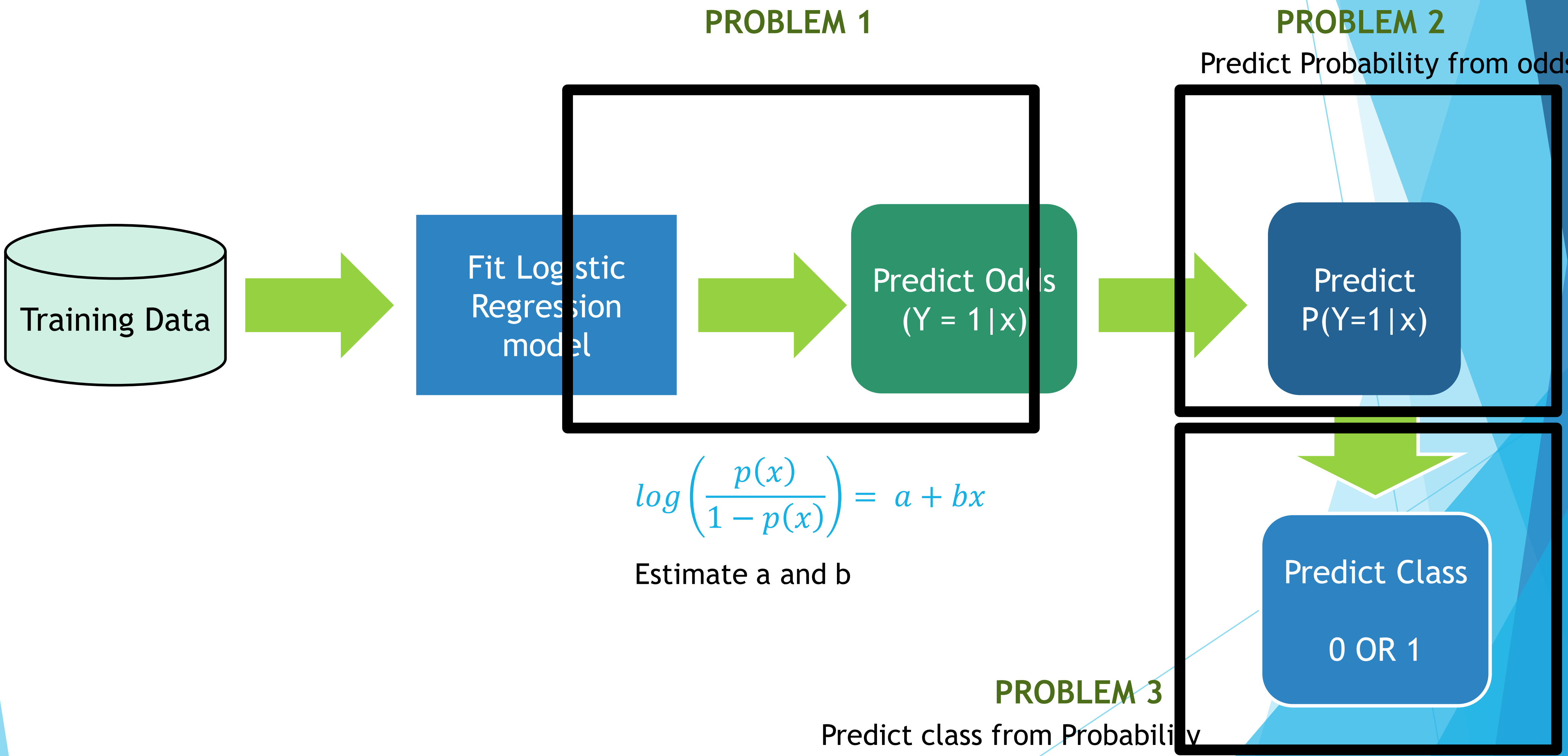
We are interested
in the Probability outcome.

Convert Odds to Probability



Our Final Interest is in Class!

How this Entire Process Work?



Problem 2 - Deriving Probability from Odds

Let, $\log(odds) = q$

$$\Longrightarrow \log\left(\frac{p}{1-p}\right) = q$$

$$\Longrightarrow \left(\frac{p}{1-p}\right) = \exp(q)$$

$$\Longrightarrow \left(\frac{p}{1-p}\right) = \mathrm{k}\,,(\mathrm{say}).$$

$$\Longrightarrow p = k(1-p)$$

$$\Longrightarrow p = k - kp$$

$$\Longrightarrow p + kp = k$$

$$\Longrightarrow \mathrm{p}(1+\mathrm{k}) = \mathrm{k}$$

$$\Longrightarrow \mathrm{p} = \frac{k}{1+k}$$

What was k? Odds right???

$Therefore,$

$$\boxed{\mathrm{p} = \frac{Odds}{1+Odds}}$$

Why Probability?

Interpretation: Probability that the tumour is malignant given that its size is 9.75 mm is 0.81 (i.e. there is 81% chance of that tumour to be malignant given this size!). The tumour is probably malignant.

Tumour — Say, Classified as 1

$P(Tumour = Malignant|size = 9.75) = 0.81$

81% Chance!

Say, Classified as 1

But how confident are we???

$P(Tumour = Malignant|size = 2.50) = 0.50$

Tumour — Say, Classified as 1

$P(Tumour = Malignant|size = 5.77) = 0.65$

Problem 3 – Predicting Class from Probability

Tumour Size	Tumour Type	$P(\widehat{Y = 1\|X} = x)$
0.5	0	0.05
2.5	0	0.12
0.75	1	0.03
5	0	0.19
15	1	0.73
⋮	⋮	⋮

Predicted Probability values

But we would like to understand…

Malignant or Benign?

Malignant or Benign?

Malignant or Benign?

Select a Threshold

Select a threshold value t such that,

$$If, \quad P(Y = 1|X = x) > t, \quad predict\ Y = 1, i.e. Malignant$$

$$If, \quad P(Y = 1|X = x) < t, \quad predict\ Y = 0, i.e. Benign$$

But what value of t should we select???

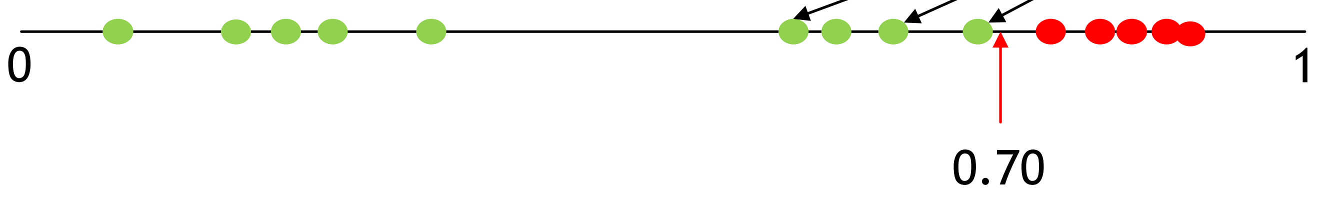
Threshold Selection

Benign

Malignant

What happens if t is large?

$P(Y = 1|X = x)$

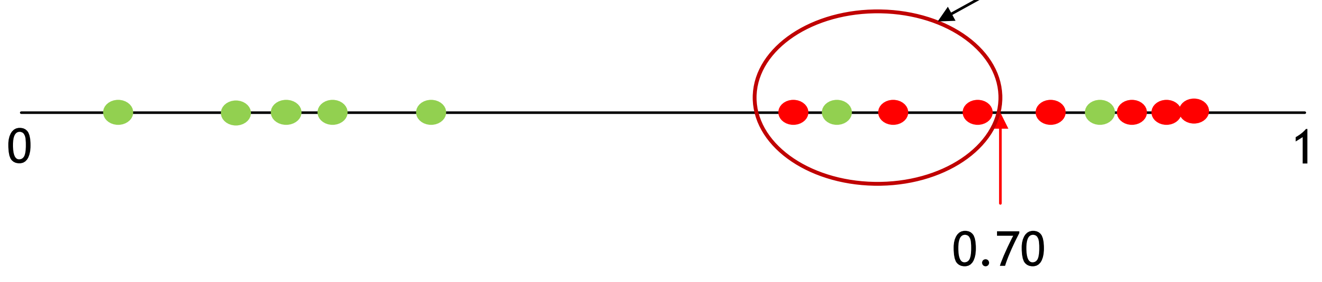


Some malignant tumours got mis-classified as benign!

Disadvantage

More misclassification here! That is we make more error in classifying malignant.

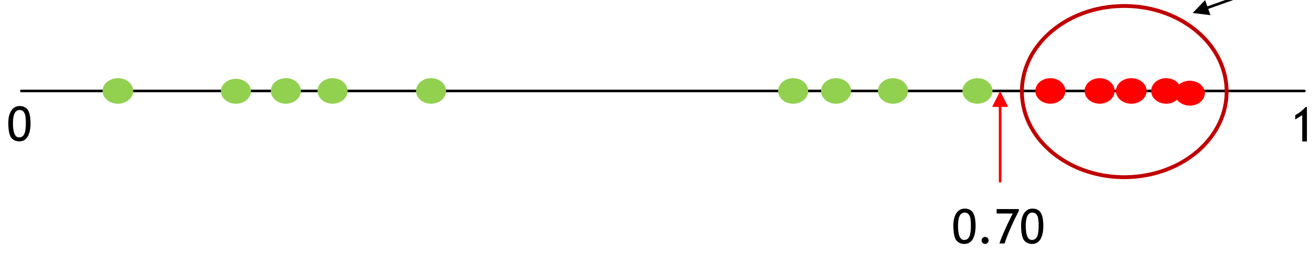
In Reality,



Threshold Selection

What happens if t is large?

$P(Y = 1|X = x)$



Advantage

Focus in detecting worst cases where immediate intervention is required (with high priority).

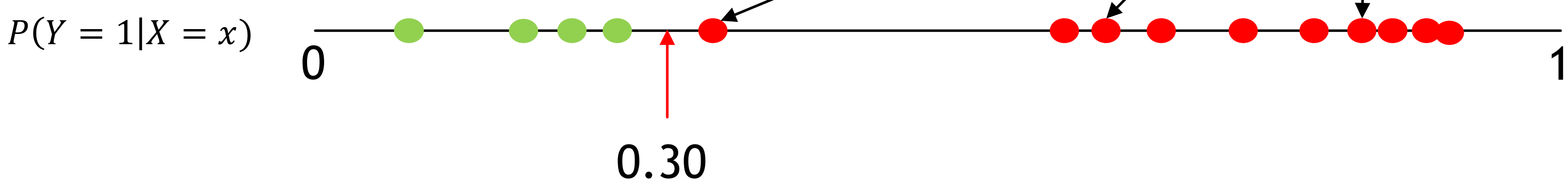
In Reality,

Threshold Selection

Disadvantage

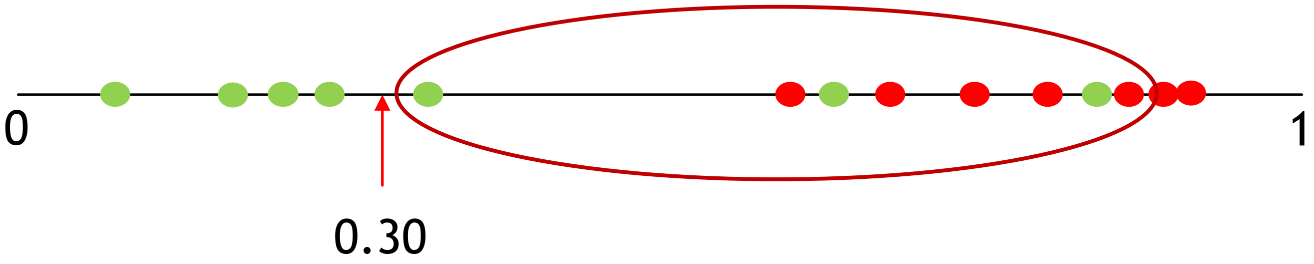
Some benign tumours got mis-classified as malignant!

What happens if t is too small?



In Reality,

More misclassification here! That is we make more error in classifying malignant.



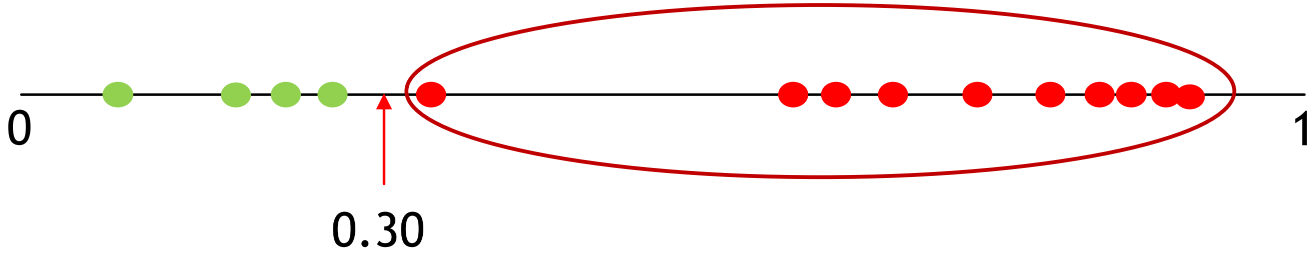
Threshold Selection

What happens if t is too small?

Advantage

Detect all patients who might be having malignant tumours and start treating them at a very early stage!

$P(Y = 1|X = x)$



In Reality,

Threshold Selection

Note:

1. Some decision-makers often have a preference for one type of error over the other, which should influence the threshold value they pick.

2. *If there's no preference between the errors, the right threshold to select is $t = 0.5$, since it just predicts the most likely outcome.*

Confusion Matrix

No. of actual positive cases that are correctly predicted as positive

No. of actual positive cases that are incorrectly predicted as negative

		Predicted Class	
		Yes	No
Actual Class	Yes	TP	FN
	No	FP	TN

No. of actual negative cases that are incorrectly predicted as positive

No. of actual negative cases that are correctly predicted as negative

Confusion Matrix: Example

	Positive (Predicted)	Negative (Predicted)
Positive (Actual)	100	50
Negative (Actual)	150	9700

1. Total number of observations = (100 + 50 + 150 + 9700) = 10000
2. Total number of positive cases = (100 + 50) = 150
3. Total number of negative cases = (150 + 9700) = 9850
4. TP = 100, i.e. 100 out of 150 positive cases are correctly predicted as positive.
5. FN = 50, i.e. 50 out of 150 positive cases are incorrectly predicted as negative.
6. FP = 150, i.e. 150 out of 9850 negative cases are incorrectly predicted as positive.
7. TN = 9700, i.e. 9700 out of 9850 negative cases are correctly predicted as negative.

Measures of Error

$$Sensitivity = \frac{TP}{TP + FN}$$

Sensitivity measures the proportion of actual positive cases that we classify correctly. This is often called the true positive rate

$$Specificity = \frac{TN}{TN + FP}$$

Specificity measures the percentage of actual negative cases that we classify correctly. This is often called the true negative rate.

$$Overall\ \mathrm{error} = \frac{FP + FN}{TP + TN + FP + FN} = \frac{\#\ mistakes}{\#\ examples}$$

Overall accuracy measures the proportion of cases that are predicted incorrectly.

Calculation of Errors

	Positive (Predicted)	Negative (Predicted)
Positive (Actual)	100	50
Negative (Actual)	150	9700

$$Sensitivity = \frac{TP}{TP + FN} = \frac{100}{100 + 50} = 0.6667$$

66.67% of the positive cases were predicted correctly by the mo

$$Specificity = \frac{9700}{9700 + 150} = 0.9847$$

98.47% of the negative cases were predicted correctly by the mo

$$Overall\ \text{error} = \frac{\#\, mistakes}{\#\, examples} = \frac{50 + 150}{10000} = 0.02$$

0.02% of examples were incorrectly predicted by the model

$$Overall\ \text{accuracy} = 1 - 0.02 = 0.98$$

Again! How to select threshold?

Some decision-makers often have a preference for one type of error over the other, which should influence the threshold value they pick.

Therefore, selection of threshold actually depends on the business problem, how much of the error can be compromised and other constraints.

ROC (Receiver Operating Characteristic) Curve